

Martin Ross

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Career Summary

- ❖ Experienced, hands-on engineer focused on building and operating production systems.
- ❖ Recognized for blending technical expertise with cross-team collaboration.
- ❖ Proven track record designing and implementing backend APIs from scratch and building responsive front-end interfaces with HTML/CSS and modern frameworks.
- ❖ Experienced in optimizing system performance, with recent focus on hardening mission-critical financial services in AWS.
- ❖ Recent roles include contract and consulting engagements focused on high-impact delivery for large and mid-size enterprise teams.

Skills / Expertise

- ❖ **Languages:** Python, TypeScript, Golang, Java, ...
- ❖ **Frameworks:** Databricks/PySpark, Node, FastAPI, Django, Express, React, Angular
- ❖ **API & Integration:** REST API design, OpenAPI, GraphQL, gRPC, WebSockets
- ❖ **Front-End Development:** HTML5, CSS3, Tailwind CSS, Responsive Design, a11y, i18n
- ❖ **Cloud & DevOps:** AWS (Lambda, Step Functions, API Gateway, Fargate), CI/CD (GitHub Actions, Jenkins, CircleCI), Terraform, Helm, Docker, Kubernetes
- ❖ **Performance, Resilience & Observability:** Instrumentation, Distributed Tracing, SLI/SLO/SLA, Profiling, Caching, Load Balancing, Chaos Engineering, OpenTelemetry, NewRelic, DataDog, ...
- ❖ **Databases:** DynamoDB, S3, MySQL, PostgreSQL, Snowflake, Redis, Cassandra
- ❖ **Other:** TDD/BDD, Agile/Scrum, Performance Optimization, Distributed Systems

Professional Experience

Capital One — San Francisco, CA | 2023 – Present

Senior Software Engineer (Contractor)

- ❖ Developed Databricks Notebooks and Spark Jobs to locate & process billions of Data Lake records.
- ❖ Led development of an Enterprise Site Reliability service combining a serverless **Golang** backend (**DynamoDB**) and a **TypeScript**/Angular frontend, delivering Error Budget dashboards, automated SLO workflows, and real-time notifications.
- ❖ Designed and implemented multiple AWS **Lambda** services in **TypeScript** and **Python** to query Snowflake datasets and trigger automated Slack alerts via **SNS** webhooks.
- ❖ Established enterprise-wide BDD integration testing patterns in **Go** and **Python**, improving test coverage and consistency across 2,000+ teams.
- ❖ Initiated and led a program leveraging **AWS FIS** and custom tooling to run controlled failure experiments in production as part of CI/CD, increasing resilience and reducing recovery times.

Crunchyroll | 2021 – 2023

Senior Software Engineer

- ❖ Developed AWS Step Function-driven **TypeScript** Lambdas for content management workflows.
- ❖ Led refactoring of four **Golang** backend services to improve testability and code coverage by 25%, increasing stability and developer productivity.
- ❖ Optimized **MySQL** queries for two high-traffic CMS services, increasing throughput by 20–50%.
- ❖ Reduced frequency and MTTR of critical incidents by 30% through code improvements, enhanced dashboards, and custom metrics with New Relic NRQL.

Independent Consultant | 2020 – 2021

Software Engineering

- ❖ Created a batch processing system using **Spring** Batch for large-scale ETL workloads.

Revinate | 2018 – 2020

Senior Software Engineer (Distributed Systems)

- ❖ Decomposed a **Java** monolith into containerized microservices on AWS Kubernetes.
- ❖ Built a high-throughput customer data pipeline using Kafka/KTable, gRPC, and Spark.
- ❖ Designed a plugin architecture enabling field technicians to deploy custom ETL logic independently, increasing core team availability by 50%.

Earlier Experience (Selected Highlights)

Navis | 2011 – 2017

Software Engineer (Backend / Integration Systems)

- ❖ Designed and implemented a Spring Boot **REST API** to integrate new crane control systems, reducing onboarding time by 75%.
- ❖ Automated CI/CD test coverage reporting and promoted TDD practices across the team.

Education

University of California, Los Angeles — B.A. in Economics